

# Jiahao Shao

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jiahao-shao1.github.io

## Education

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**The Hong Kong University of Science and Technology**

2026 – Present

Ph.D. in Computer Science

**Zhejiang University**

2020 – 2024

B.E. in Automation

## Experiences

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**Ant Group**

2024 – 2026

Research Intern

Advisor: Yujun Shen

**Zhejiang University**

2023 – Present

Research Assistant

Advisor: Yiyi Liao

## Research Interests

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My long-term research goal is to build a general-purpose world model that can understand, simulate, and interact with the physical world. My research interests lie at the intersection of Computer Vision, Generative Modeling, and Embodied AI.

## Publications

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(\* Equal Contribution; † Equal Advising)

- [1] [Jiahao Shao](#), Yuanbo Yang, Yiyi Liao, Yujun Shen, Ceyuan Yang†, Yinghao Xu†, “Thinking With Tools, Not With Pixels: Tool Calls as Text Scaffolds for Visual Reasoning”, *Under review*, 2026.
- [2] Yuanbo Yang, [Jiahao Shao](#), Yiyi Liao, Jun Gao, “From a Spiral to a Sink: How Positional Encoding Shapes Attention Sinks in Diffusion Transformers”, *Under review*, 2026.
- [3] Zhen Xu\*, Hongyu Zhou\*, Sida Peng, Haotong Lin, Haoyu Guo, [Jiahao Shao](#), Peishan Yang, Qinglin Yang, Sheng Miao, Xingyi He, Yifan Wang, Yue Wang, Ruizhen Hu, Yiyi Liao, Xiaowei Zhou, Hujun Bao, “Towards Depth Foundation Model: Recent Trends in Vision-Based Depth Estimation”, *Computational Visual Media*, 2026.
- [4] Chunyang Li\*, Yuanbo Yang\*, [Jiahao Shao](#)\*, Hongyu Zhou, Katja Schwarz, Yiyi Liao, “ReRoPE: Repurposing RoPE for Relative Camera Control”, *SIGGRAPH*, 2026.

- [5] Jiabao Wang, Hongyu Zhou, Yuanbo Yang, [Jiahao Shao](#), Yiyi Liao, “The Constant Eye: Benchmarking and Bridging Appearance Robustness in Autonomous Driving”, arXiv preprint, 2026.
- [6] Hanlei Guo, [Jiahao Shao](#), Xinya Chen, Xiyang Tan, Sheng Miao, Yujun Shen, Yiyi Liao, “ScenDi: 3D-to-2D Scene Diffusion Cascades for Urban Generation”, *Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [7] Yuanbo Yang\*, [Jiahao Shao](#)\*, Xinyang Li, Yujun Shen, Andreas Geiger, Yiyi Liao, “Prometheus: 3D-Aware Latent Diffusion Models for Feed-Forward Text-to-3D Scene Generation”, *Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [8] [Jiahao Shao](#)\*, Yuanbo Yang\*, Hongyu Zhou, Youmin Zhang, Yujun Shen, Vitor Guizilini, Yue Wang, Matteo Poggi, Yiyi Liao, “Learning Temporally Consistent Video Depth from Video Diffusion Priors”, *Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [9] Hongyu Zhou, [Jiahao Shao](#), Lu Xu, Dongfeng Bai, Weichao Qiu, Bingbing Liu, Yue Wang, Andreas Geiger, Yiyi Liao, “HUGS: Holistic Urban 3D Scene Understanding via Gaussian Splatting”, *Computer Vision and Pattern Recognition (CVPR)*, 2024.

## Honors and Awards

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|-----------------------------------------------------------|-------------|
| Finalist Award, Mathematical Contest in Modeling (top 2%) | 2023        |
| Second-prize Scholarship of Zhejiang University           | 2020 – 2022 |

## Open Source Projects

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### **sjh-skills**

A collection of Claude Code skills for research workflow automation.

## Skills

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Python, C, PyTorch, PyTorch Lightning, MATLAB,  $\LaTeX$ , Linux